

QA. Given two queries P and Q that give the same result, explain which one you would choose for efficient query processing and why?

1) **P:** select subject, count(subject)
from student_details
where subject != 'Science'
and subject != 'Maths'
group by subject;

Q: select subject, count(subject)
from student_details
group by subject
having subject!= 'Science' and
subject != 'Maths';

2) **P:** select name
from employee
where salary = (select
max(salary) from
employee_details)
and age = (select max(age)
from employee_details)
and emp_dept = 'Electronics';

Q: select name
from employee
where (salary, age) =
(select max (salary), max (age)
from employee_details)
and dept = 'Electronics';

3) **P:** select id, name, salary
from employee
where salary < 25000;

Q: select id, name, salary
from employee
where salary + 10000 < 35000;

4) **P:** select D.cust_name from Branch
B, Accounts A, Depositor D where
B.branchname=A.branchname and
A.accountno =D.accountno and
B.branchcity="Mumbai";

Q: select D.cust_name from Accounts A,
Depositor D where A.accountno =
D.accountno and A.branchname in (select
B.branchname from Accounts B, R where
B.branchname in (select R.branchname from
Branch where branchcity="Mumbai") R);

QB. Find the number of disk accesses required for each of the following queries. Assume T(Staff)=1000, T(Branch)=50, 50 managers (one for each branch) and 5 Vasco branches. Also assume there are no indexes or sort keys on both relation, and that results of any intermediate operations are stored on disk. Ignore cost of final write as it is same in each case. The tuples are accessed one at a time and main memory is large enough to process entire relations.

1. $\sigma_{(\text{position}='Manager') \wedge (\text{city}='Vasco') \wedge (\text{staff} . \text{branchno} = \text{Branch} . \text{branchno})} (\text{Staff} \times \text{Branch})$
2. $\sigma_{(\text{position}='Manager') \wedge (\text{city}='Vasco')} (\text{Staff} \bowtie_{\text{staff} . \text{branchno} = \text{Branch} . \text{branchno}} \text{Branch})$
3. $(\sigma_{\text{position}='Manager'} \text{Staff}) \bowtie_{\text{staff} . \text{branchno} = \text{Branch} . \text{branchno}} (\sigma_{\text{city}='Vasco'} \text{Branch})$

QC. The vital statistics for four relations W, X, Y, Z are given below. Estimate the sizes of relations that are results of the following expressions.

W(a,b)	X(b,c)	Y(c,d)	Z(d,e)
T(W) = 100	T(X) = 200	T(Y) = 300	T(Z) = 400
V(W,a) = 20	V(X,b) = 50	V(Y,c) = 50	V(Z,d) = 40
V(W,b) = 60	V(X,c) = 100	V(Y,d) = 50	V(Z,d) = 100

a) $W \bowtie X \bowtie Y \bowtie Z$

b) $\sigma_{a=10}(W)$

c) $\sigma_{c=20}(Y) \bowtie Z$

d) $\sigma_{b=10 \wedge c>20}(X)$